**Project Design Phase**
**Problem – Solution Fit Template**

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| Date | 22 February 2026 |
| Team ID | LTVIP2026TMIDS42033 |
| Project Name | **Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management** |
| Maximum Marks | 2 Marks |

**Problem – Solution Fit Template:**

The Problem–Solution Fit for this project focuses on addressing the challenges faced by wind farm operators and grid managers in accurately forecasting wind turbine energy output. Fluctuating weather conditions, unpredictable wind speeds, and the lack of real-time predictive tools lead to grid imbalance, inefficient energy distribution, and reduced renewable energy performance.

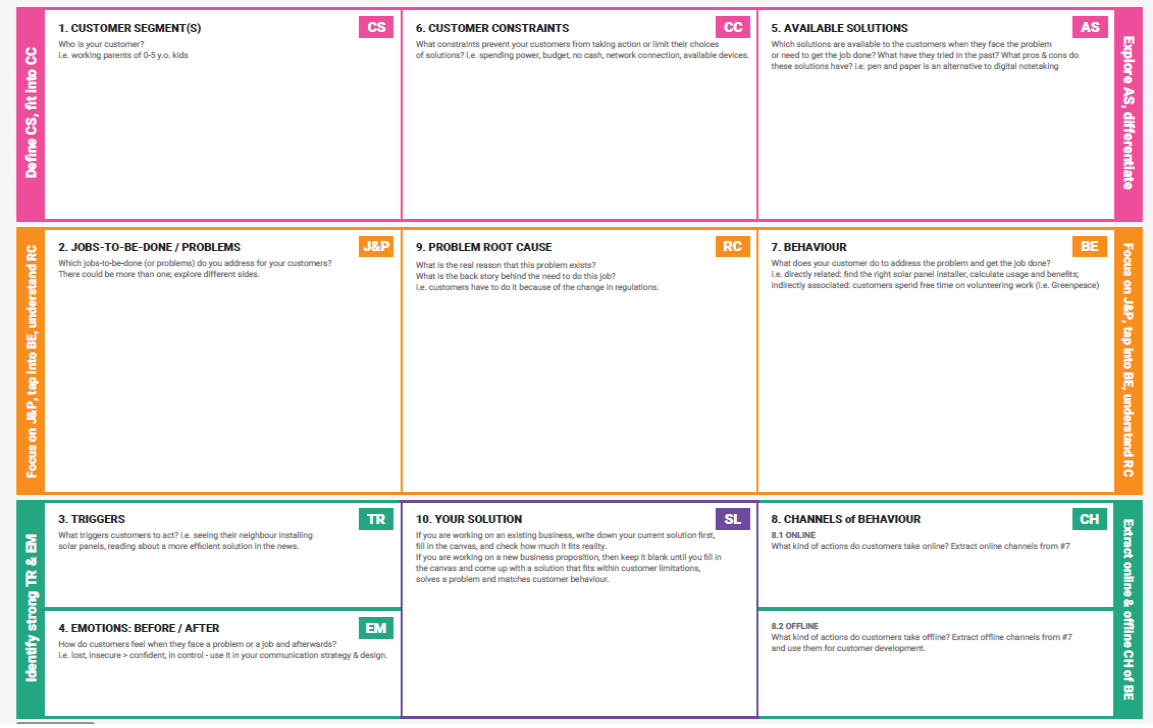
This solution integrates real-time weather data with machine learning models to predict wind turbine energy output accurately. By providing an intelligent, data-driven dashboard, the system helps decision-makers optimize turbine performance, enhance grid stability, and improve renewable energy management.

**Purpose:**

- Solve the problem of unpredictable wind energy output using AI-based forecasting.
- Enable faster and more accurate decision-making for grid operators.
- Reduce power imbalance and improve renewable energy efficiency.
- Provide real-time monitoring and predictive analytics through an interactive dashboard.
- Enhance turbine performance optimization using data-driven insights.

Support sustainable energy management with scalable and modular architecture

**Template:**



## Target Customers:

- Wind Farm Operators
- Grid Managers
- Renewable Energy Companies
- Energy Distribution Authorities
- Smart Grid System Administrators

## Identified Problems:

- Unpredictable wind patterns affecting energy generation.
- Difficulty in accurate real-time energy forecasting.
- Grid instability due to fluctuating renewable supply.
- Inefficient turbine utilization.
- Lack of centralized AI-powered monitoring tools.

## Proposed Solution:

- Integration of OpenWeather API for real-time weather data.
- Machine Learning model for wind turbine energy prediction.
- Flask-based backend for processing and model integration.
- React-based interactive dashboard for visualization.
- Energy analytics with gauges, charts, and prediction history tracking.

## Value Proposition:

The proposed system delivers accurate, real-time wind energy predictions that improve grid stability, enhance renewable energy utilization, and support smarter energy management decisions. It reduces operational risks, increases efficiency, and provides scalable architecture for future expansion.

## References:


1. https://c4model.com/
2. https://openweathermap.org/api
3. https://scikit-learn.org/
4. https://flask.palletsprojects.com/
5. https://react.dev/
6. https://aws.amazon.com/architecture/